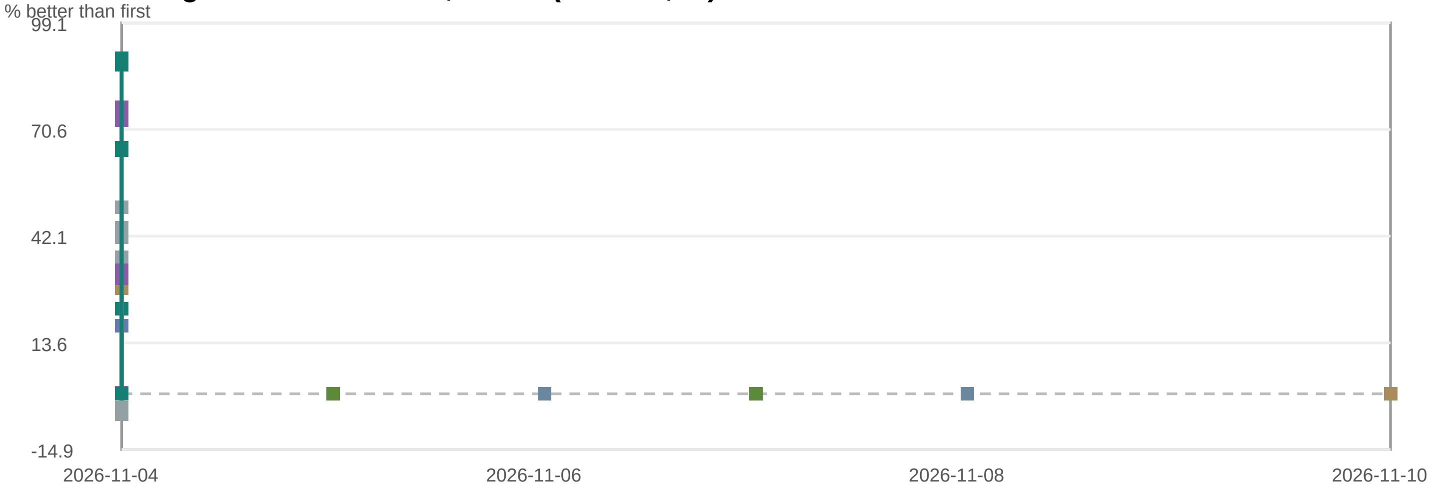
# Benchmark progress: crypto

From bench.record/v1 records in Git; raw data in S3 by SHA-256. Derived numbers are labelled derived.

Records: open 3, measured 80, invalid 0, censored 3. Headline unit S: group operations / sqrt(n); the ECDLP cost unit, every phase priced.

## Cumulative gain vs first record, unit S (derived, %)




### Series key

archive:ic-packed-filter-p32-2
archive:ic-packed-filter-p32-2
dlpwai:static-dh-p224-shape •
dlpwai:static-dh-p224-shape •
dlpwai:static-dh-p224-shape •
ecdlp:curve25519-family-k20 •
ecdlp:curve25519-family-k24 •
ecdlp:curve25519-family-k28 •
ecdlp:curve25519-family-k32 •
ic:curve25519-family-k20-32 •
ic:curve25519-family-k20-32 •
ic:curve25519-family-k20-32 •
ic:curve25519-family-k20-32 •
ic:curve25519-family-k20-32 •
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ic:curve25519-family-k20-32 •
ic:curve25519-family-k20-32 •
ic:curve25519-family-k20-32 •
ic:curve25519-family-k20 • s_w
ic:curve25519-family-k24-28 •
ic:curve25519-family-k24-28 •
ic:curve25519-family-k24-32 •
ic:curve25519-family-k24-32 •
ic:curve25519-family-k24-32 •
ic:curve25519-family-k24-32 •
ic:curve25519-family-k24-32 •
ic:curve25519-family-k24 • s_w
ic:curve25519-family-k28-32 •
ic:curve25519-family-k28-32 •
ic:curve25519-family-k28-32 •
ic:curve25519-family-k28 • s_w
ic:curve25519-family-k32 • s_w
ic:curve25519-k20-single-targe
ic:curve25519-k20-single-targe
ic:curve25519-k20-single-targe
ic:curve25519-k20-single-targe
ic:curve25519-k20-single-targe
ic:curve25519-k20-single-targe
ic:curve25519-k20-single-targe
ic:curve25519-k24-single-targe
ic:curve25519-k24-single-targe
ic:curve25519-k24-single-targe
ic:curve25519-k24-single-targe
ic:curve25519-k24-single-targe
ic:curve25519-k24-single-targe
ic:curve25519-k24-single-targe
kernel:c25519_family_bsgs_vs_r
kernel:c25519_family_bsgs_vs_r
rho:prime_medium_tier_bound •
rho:prime_medium_tier_bound •
rho:prime_medium_tier_bound •

### Cumulative gain vs first record, unit S (derived, %) (series key continued)

- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
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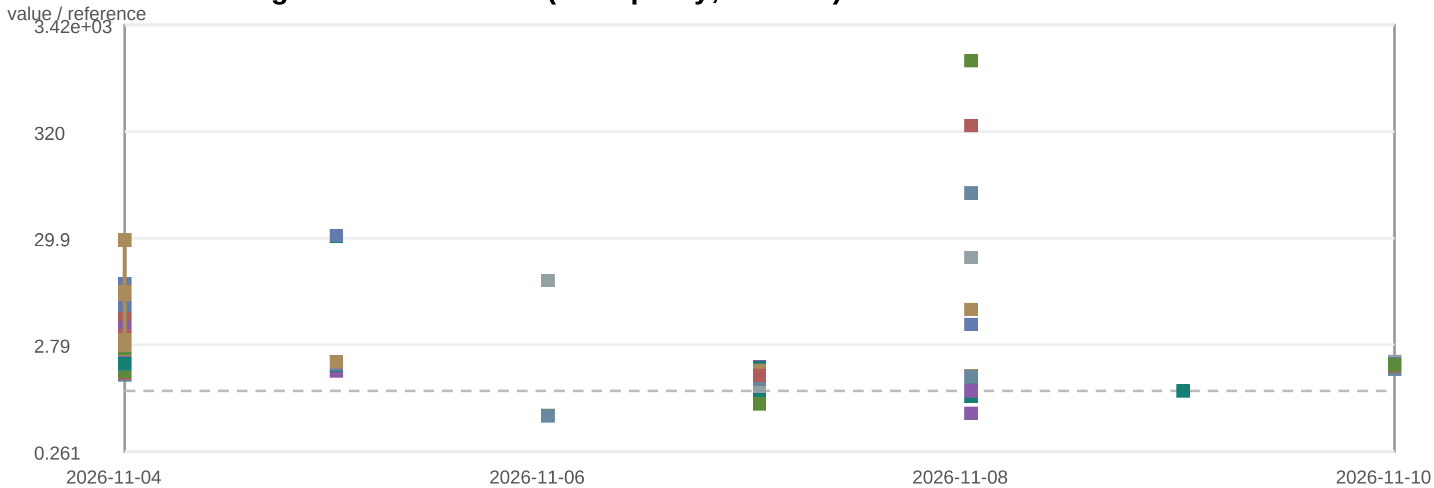
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •

## Distance remaining: value / reference (1.0 = parity; derived)




### Series key

- archive:ic-packed-filter-p32-2
- archive:ic-packed-filter-p32-2
- dlpwai:static-dh-p224-shape •
- dlpwai:static-dh-p224-shape •
- dlpwai:static-dh-p224-shape •
- ecdlp:curve25519-family-k20 •
- ecdlp:curve25519-family-k20 •
- ecdlp:curve25519-family-k24 •
- ecdlp:curve25519-family-k24 •
- ecdlp:curve25519-family-k28 •
- ecdlp:curve25519-family-k28 •
- ecdlp:curve25519-family-k32 •
- ecdlp:curve25519-family-k32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
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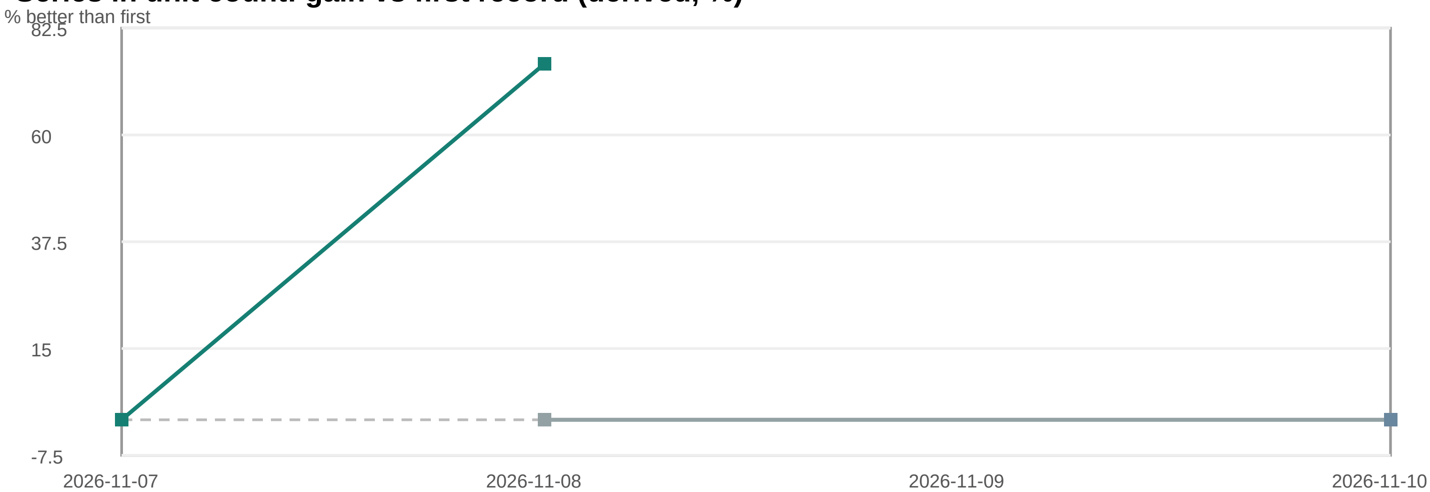
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20 • s_w
- ic:curve25519-family-k24-28 •
- ic:curve25519-family-k24-28 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24 • s_w
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28 • s_w
- ic:curve25519-family-k32 • s_w
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe

**Distance remaining: value / reference (1.0 = parity; derived) (series key continued)**

- ic:curve25519-k24-single-targe
- kernel:c25519_family_bsgs_vs_r
- kernel:c25519_family_bsgs_vs_r
- study:minors_guess_determine_2
- study:minors_guess_determine_2
- study:minors_guess_determine_2
- study:minors_guess_determine_2
- study:minors_guess_determine_2
- study:minors_guess_determine_2
- study:pkm_falldegree_unblock_2
- study:pkm_falldegree_unblock_2

## Series in unit count: gain vs first record (derived, %)




**Series key**

- analysis:ecbench_statistics_v2
- archive:icb-large-prime-n53-s1
- archive:icb-large-prime-n53-s2
- archive:icb-large-prime-n61-s1
- archive:icb-large-prime-n61-s2
- ecdlp:split-prime-p3-cross-tra
- ecdlp:split-prime-p3-cross-tra
- ic:koblitz-chain-s3-arity-m5m6
- ic:split-prime-p1-rr-dickson-t
- ic:split-prime-p1-rr-dickson-t
- ic:split-prime-p1-rr-dickson-t
- ic:split-prime-p1-rr-dickson-t
- ic:split-prime-p1-rr-dickson-t
- ic:split-prime-p2-twist-coset-
- ic:split-prime-p2-twist-coset-
- registry-rebinding:ic-aimed-ga
- study:elliptic_code_list_decod
- study:pkm_falldegree_unblock_2
- study:pkm_falldegree_unblock_2
- study:pkm_falldegree_unblock_2
- study:pkm_falldegree_unblock_2
- study:pkm_falldegree_unblock_2
- study:pkm_falldegree_unblock_2
- study:pkm_falldegree_unblock_2
- study:pkm_falldegree_unblock_2
- study:pkm_falldegree_unblock_2
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- study:pkm_falldegree_unblock_2
- study:pkm_falldegree_unblock_2
- study:pkm_falldegree_unblock_2
- study:pkm_falldegree_unblock_2
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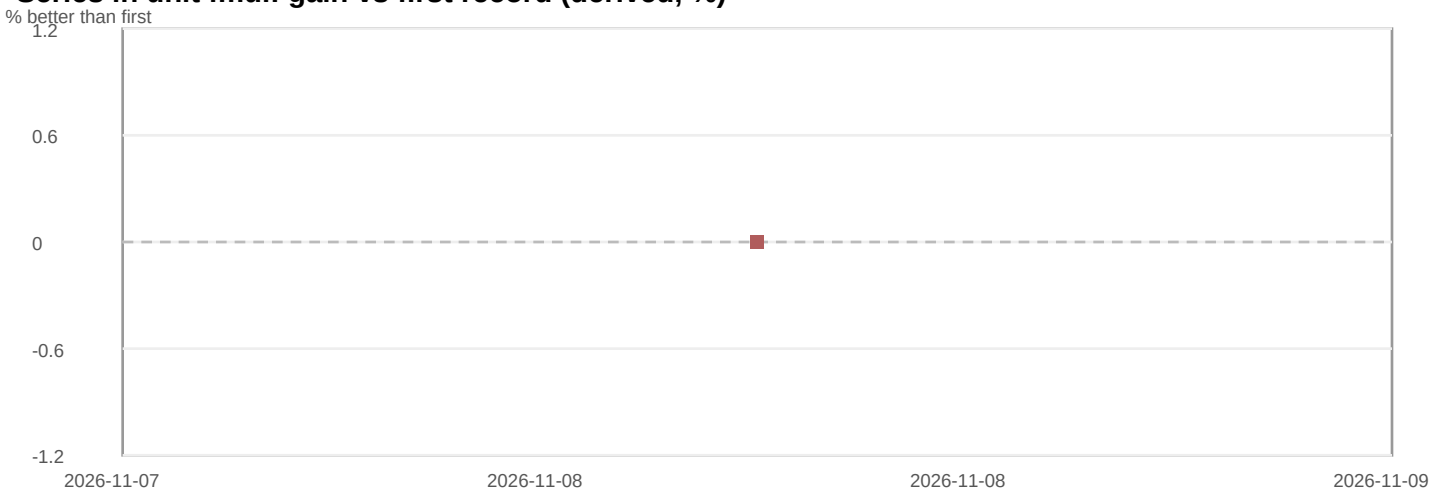
- study:pkm_falldegree_unblock_2
- study:pkm_falldegree_unblock_2
- study:pkm_falldegree_unblock_2
- study:pkm_falldegree_unblock_2

## Series in unit fmul: gain vs first record (derived, %)




### Series key

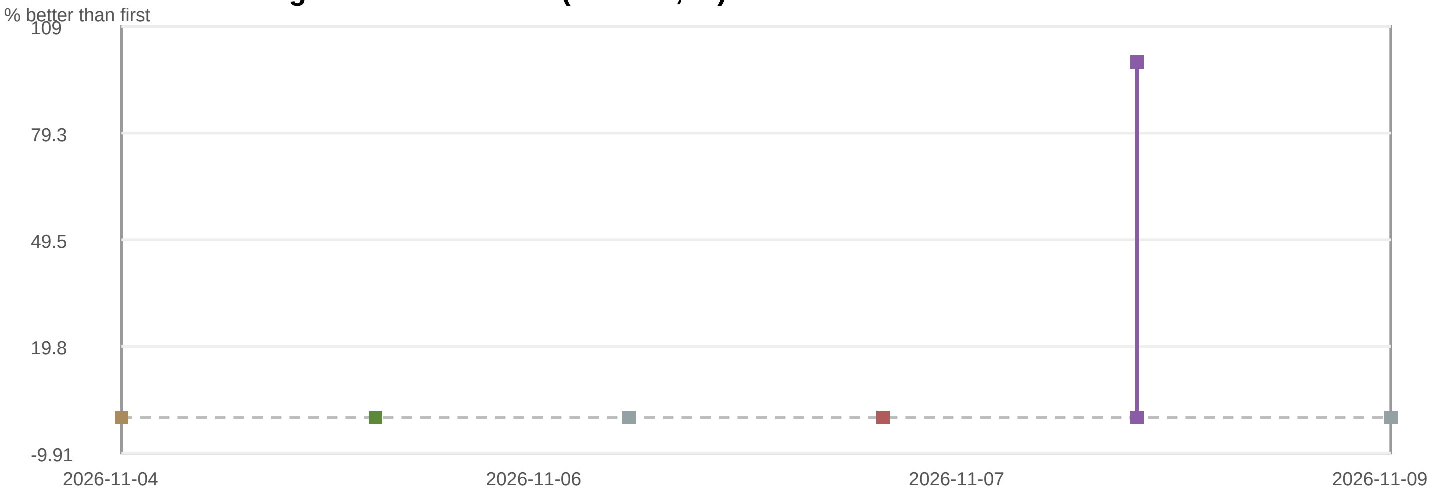
- study:minors_guess_determine_2
- study:minors_guess_determine_2
- study:minors_guess_determine_2
- study:minors_guess_determine_2
- study:minors_guess_determine_2
- study:minors_guess_determine_2
- study:pkm_falldegree_unblock_2
- study:pkm_falldegree_unblock_2

## Series in unit percent: gain vs first record (derived, %)

### Series key

- ecdlp:split-prime-p3-cross-tra
- ic:split-prime-p2-twist-coset-
- ic:split-prime-p2-twist-coset-
- ic:split-prime-p2-twist-coset-

## Series in unit ratio: gain vs first record (derived, %)




### Series key

- analysis:ecbench_statistics_v2
- analysis:ecbench_statistics_v2
- archive:ic-packed-filter-p32-2
- ecdlp:curve25519-family-k20 •
- ecdlp:curve25519-family-k20 •
- ecdlp:curve25519-family-k20 •
- ecdlp:curve25519-family-k24 •
- ecdlp:curve25519-family-k24 •
- ecdlp:curve25519-family-k24 •
- ecdlp:curve25519-family-k28 •
- ecdlp:curve25519-family-k28 •
- ecdlp:curve25519-family-k28 •
- ecdlp:curve25519-family-k32 •
- ecdlp:curve25519-family-k32 •
- ecdlp:curve25519-family-k32 •
- ecdlp:split-prime-p3-cross-tra
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
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- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:split-prime-p1-rr-dickson-t
- ic:split-prime-p1-rr-dickson-t
- ic:split-prime-p2-twist-coset-
- ic:split-prime-p2-twist-coset-
- kernel:f6_ic_gate • f4_control
- kernel:f6_ic_gate • f6ic_cpu_o
- kernel:f6_ic_gate • f6ic_cpu_o
- kernel:f6_ic_gate • f6ic_word_
- kernel:inherited_f4_solve_tree
- kernel:inherited_f4_solve_tree
- kernel:inherited_f4_solve_tree
- kernel:inherited_f4_solve_tree
- rho:koblitz_signed_frobenius_p
- rho:koblitz_signed_frobenius_p
- rho:koblitz_signed_frobenius_p
- rho:koblitz_signed_frobenius_p
- rho:koblitz_signed_frobenius_p
- rho:prime_medium_p4_confirm •
- rho:prime_medium_p4_confirm •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- study:ml_learnability_control_
- study:ml_learnability_control_
- study:ml_learnability_control_
- study:ml_learnability_control_
- study:ml_learnability_control_
- study:ml_learnability_control_
- study:ml_learnability_control_
- study:ml_learnability_control_
- study:ml_learnability_control_
- study:ml_learnability_control_

**Series in unit ratio: gain vs first record (derived, %) (series key continued)**

- study:ml_learnability_control_
- study:ml_learnability_control_
- study:ml_learnability_control_
- study:ml_learnability_control_
- study:ml_learnability_control_
- study:ml_learnability_control_
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- study:ml_learnability_control_
- study:ml_learnability_control_
- study:ml_learnability_control_
- study:ml_learnability_control_
- study:ml_learnability_control_
- study:ml_learnability_control_
- study:split_graph_secant_cost_
- study:split_graph_secant_cost_
- study:split_graph_secant_cost_
- study:split_graph_secant_cost_
- study:split_graph_secant_cost_
- study:split_graph_secant_cost_
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- study:split_graph_secant_cost_
- study:split_graph_secant_cost_
- study:split_graph_secant_cost_

## Series summary

| problem • metric | unit | first | latest | gain vs first | value/ref | pts |
|---|---|---|---|---|---|---|
| analysis:ecbench_statistics_v2 • censored_ratio_ | ratio | 648.2 | 648.2 | - | - | 1 |
| analysis:ecbench_statistics_v2 • heavy_tail_rows | count | 71 | 71 | - | - | 1 |
| analysis:ecbench_statistics_v2 • ops_interval_wi | ratio | 13.1 | 13.1 | - | - | 1 |
| archive:ic-packed-filter-p32-20261029 • s_prime_ | S | 31.6936 | 31.6936 | 0% | 31.4718 | 1 |
| archive:ic-packed-filter-p32-20261029 • s_prime_ | S | 63.4961 | 63.4961 | 0% | 31.6255 | 1 |
| archive:ic-packed-filter-p32-20261029 • wall_per | ratio | 0.9601 | 0.9601 | 0% | - | 1 |
| archive:icb-large-prime-n53-s1-20261029 • report | count | 240 | 240 | - | - | 1 |
| archive:icb-large-prime-n53-s2-20261029 • report | count | 240 | 240 | - | - | 1 |
| archive:icb-large-prime-n61-s1-20261029 • report | count | 72 | 72 | - | - | 1 |
| archive:icb-large-prime-n61-s2-20261029 • report | count | 60 | 60 | - | - | 1 |
| dlpwai:static-dh-p224-shape • s_w0_cheon_pminus1 | S | 0.5739 | 0.5739 | 0% | 0.5814 | 2 |
| dlpwai:static-dh-p224-shape • s_w0_cheon_pminus1 | S | 12.2426 | 12.2426 | 0% | 11.6066 | 2 |
| dlpwai:static-dh-p224-shape • s_w0_cheon_pplus1_ | S | 0.5819 | 0.5819 | 0% | 0.5746 | 2 |
| ecdlp:curve25519-family-k20 • ratio_j16_over_rho | ratio | 1.3581 | 1.3581 | 0% | 1.3581 | 1 |
| ecdlp:curve25519-family-k20 • ratio_j32_over_j16 | ratio | 1.0507 | 1.0507 | 0% | - | 1 |
| ecdlp:curve25519-family-k20 • ratio_j4_over_j16_ | ratio | 1.2204 | 1.2204 | 0% | - | 1 |
| ecdlp:curve25519-family-k20 • s_twcj_j16_w0 | S | 2.1364 | 2.1364 | 0% | 1.3581 | 1 |
| ecdlp:curve25519-family-k24 • ratio_j16_over_rho | ratio | 1.3431 | 1.3431 | 0% | 1.3431 | 1 |
| ecdlp:curve25519-family-k24 • ratio_j32_over_j16 | ratio | 1.0141 | 1.0141 | 0% | - | 1 |
| ecdlp:curve25519-family-k24 • ratio_j4_over_j16_ | ratio | 1.2682 | 1.2682 | 0% | - | 1 |
| ecdlp:curve25519-family-k24 • s_twcj_j16_w0 | S | 1.5834 | 1.5834 | 0% | 1.3427 | 1 |
| ecdlp:curve25519-family-k28 • ratio_j32_over_j16 | ratio | 0.9966 | 0.9966 | 0% | - | 1 |
| ecdlp:curve25519-family-k28 • ratio_twcj_over_f1 | ratio | 0.9059 | 0.9059 | 0% | - | 1 |
| ecdlp:curve25519-family-k28 • ratio_twcj_over_rh | ratio | 1.3951 | 1.3951 | 0% | 1.3951 | 1 |
| ecdlp:curve25519-family-k28 • s_twcj_j16_w0 | S | 1.4349 | 1.4349 | 0% | 1.395 | 1 |
| ecdlp:curve25519-family-k32 • ratio_j32_over_j16 | ratio | 0.9853 | 0.9853 | 0% | - | 1 |
| ecdlp:curve25519-family-k32 • ratio_twcj_over_f1 | ratio | 0.8569 | 0.8569 | 0% | - | 1 |
| ecdlp:curve25519-family-k32 • ratio_twcj_over_rh | ratio | 1.3663 | 1.3663 | 0% | 1.3663 | 1 |
| ecdlp:curve25519-family-k32 • s_twcj_j16_w0 | S | 1.3871 | 1.3871 | 0% | 1.3663 | 1 |
| ecdlp:split-prime-p3-cross-trace-census • aa_rho | ratio | 0.986 | 0.009 | 99.0872% | - | 2 |
| ecdlp:split-prime-p3-cross-trace-census • bonfer | count | 0 | 0 | - | - | 1 |
| ecdlp:split-prime-p3-cross-trace-census • bonfer | count | 0 | 0 | - | - | 1 |
| ecdlp:split-prime-p3-cross-trace-census • max_ab | percent | 0.7 | 0.087 | 87.5714% | - | 2 |
| ic:curve25519-family-k20-32 • c1confirmk20 | S | 2.227 | 2.227 | 0% | 1.4006 | 1 |
| ic:curve25519-family-k20-32 • c1confirmk24 | S | 1.814 | 1.814 | 0% | 1.5898 | 1 |
| ic:curve25519-family-k20-32 • c1confirmk28 | S | 1.711 | 1.711 | 0% | 1.7127 | 1 |
| ic:curve25519-family-k20-32 • c1confirmk32 | S | 1.545 | 1.545 | 0% | 1.6649 | 1 |
| ic:curve25519-family-k20-32 • inclazyk24 | S | 2.096 | 2.096 | 0% | 1.7838 | 1 |
| ic:curve25519-family-k20-32 • inclazyk28 | S | 1.932 | 1.932 | 0% | 1.9148 | 1 |
| ic:curve25519-family-k20-32 • inclazyk32 | S | 1.908 | 1.908 | 0% | 1.9292 | 1 |
| ic:curve25519-family-k20-32 • r36selk20 | S | 2.446 | 2.446 | 0% | 1.6394 | 1 |
| ic:curve25519-family-k20-32 • r36selk24 | S | 2.016 | 2.016 | 0% | 1.792 | 1 |
| ic:curve25519-family-k20-32 • r36selk28 | S | 1.768 | 1.768 | 0% | 1.7402 | 1 |
| ic:curve25519-family-k20-32 • r36selk32 | S | 1.772 | 1.772 | 0% | 1.7632 | 1 |
| ic:curve25519-family-k20-32 • r37selk20 | S | 2.469 | 2.469 | 0% | 1.6053 | 1 |
| ic:curve25519-family-k20-32 • r37selk28 | S | 1.834 | 1.834 | 0% | 1.8249 | 1 |
| ic:curve25519-family-k20-32 • r37selk32 | S | 1.715 | 1.715 | 0% | 1.7754 | 1 |
| ic:curve25519-family-k20-32 • ratio_w0_k24 | ratio | 0.9375 | 0.9375 | 0% | - | 1 |
| ic:curve25519-family-k20-32 • ratio_w0_k32 | ratio | 0.8802 | 0.8802 | 0% | - | 1 |
| ic:curve25519-family-k20-32 • s_w0_incumbent_k20 | S | 2.5746 | 2.5746 | 0% | 1.7601 | 1 |
| ic:curve25519-family-k20-32 • s_w0_incumbent_k24 | S | 2.1638 | 1.6735 | 22.6592% | 1.429 | 2 |
| ic:curve25519-family-k20-32 • s_w0_incumbent_k28 | S | 1.9211 | 1.5717 | 18.1849% | 1.4943 | 2 |
| ic:curve25519-family-k20-32 • s_w0_incumbent_k32 | S | 2.11 | 1.5152 | 28.1881% | 1.5539 | 2 |
| ic:curve25519-family-k20-32 • s_w0_mean5curves_k | S | 2.601 | 2.601 | 0% | 1.5669 | 1 |
| ic:curve25519-family-k20-32 • s_w0_mean5curves_k | S | 2.111 | 2.111 | 0% | 1.7695 | 1 |

| | | | | | | |
|---|---|---|---|---|---|---|
| ic:curve25519-family-k20-32 • s_w0_mean5curves_k | S | 1.869 | 1.869 | 0% | 1.7971 | 1 |
| ic:curve25519-family-k20-32 • s_w0_mean5curves_k | S | 1.93 | 1.93 | 0% | 1.9052 | 1 |
| ic:curve25519-family-k20-32 • tw2cjlazyk20 | S | 2.529 | 2.529 | 0% | 1.6108 | 1 |
| ic:curve25519-family-k20-32 • tw2cjlazyk24 | S | 1.915 | 1.915 | 0% | 1.6298 | 1 |
| ic:curve25519-family-k20-32 • tw2cjlazyk28 | S | 1.756 | 1.756 | 0% | 1.7403 | 1 |
| ic:curve25519-family-k20-32 • tw2cjlazyk32 | S | 1.758 | 1.758 | 0% | 1.7776 | 1 |
| ic:curve25519-family-k20 • s_w0_incumbent | S | 8.3478 | 4.8598 | 41.7827% | 3.2356 | 5 |
| ic:curve25519-family-k24-28 • s_w0_incumbent_k24 | S | 2.5842 | 2.5842 | 0% | 2.0304 | 1 |
| ic:curve25519-family-k24-28 • s_w0_incumbent_k28 | S | 2.3732 | 2.3732 | 0% | 2.3865 | 1 |
| ic:curve25519-family-k24-32 • best_train_s_w0_k2 | S | 2.713 | 2.713 | 0% | - | 1 |
| ic:curve25519-family-k24-32 • best_train_s_w0_k3 | S | 2.768 | 2.768 | 0% | - | 1 |
| ic:curve25519-family-k24-32 • ratio_w0_k32 | ratio | 0.9106 | 0.9106 | 0% | - | 1 |
| ic:curve25519-family-k24-32 • s_w0_seedrun_k24 | S | 2.1055 | 2.1055 | 0% | 1.942 | 1 |
| ic:curve25519-family-k24-32 • s_w0_seedrun_k28 | S | 1.8835 | 1.8835 | 0% | 1.8154 | 1 |
| ic:curve25519-family-k24-32 • s_w0_seedrun_k32 | S | 1.8423 | 1.8423 | 0% | 1.8232 | 1 |
| ic:curve25519-family-k24 • s_w0_incumbent | S | 6.607 | 3.3182 | 49.7782% | 2.7163 | 5 |
| ic:curve25519-family-k28-32 • s_w0_incumbent | S | 2.5798 | 2.5798 | 0% | 2.5891 | 1 |
| ic:curve25519-family-k28-32 • s_w0_seed_mean_k28 | S | 1.969 | 1.969 | 0% | 1.8593 | 1 |
| ic:curve25519-family-k28-32 • s_w0_seed_mean_k32 | S | 1.781 | 1.781 | 0% | 1.8229 | 1 |
| ic:curve25519-family-k28 • s_w0_incumbent | S | 10.9395 | 2.576 | 76.452% | 2.7572 | 7 |
| ic:curve25519-family-k32 • s_w0_incumbent | S | 24.1377 | 2.519 | 89.5641% | 2.7453 | 5 |
| ic:curve25519-k20-single-target • ratio_f2_over_ | ratio | 1.2283 | 1.2283 | -0% | - | 1 |
| ic:curve25519-k20-single-target • ratio_f2_over_ | ratio | 1.2385 | 1.2385 | -0% | - | 1 |
| ic:curve25519-k20-single-target • ratio_f2_over_ | ratio | 1.1542 | 1.1542 | -0% | - | 1 |
| ic:curve25519-k20-single-target • ratio_tw_over_ | ratio | 1.1656 | 1.1656 | 0% | - | 1 |
| ic:curve25519-k20-single-target • ratio_twcj_ove | ratio | 0.8872 | 0.8872 | 0% | - | 1 |
| ic:curve25519-k20-single-target • ratio_twcj_ove | ratio | 0.9088 | 0.9088 | 0% | - | 1 |
| ic:curve25519-k20-single-target • ratio_twcj_ove | ratio | 0.7335 | 0.7335 | 0% | - | 1 |
| ic:curve25519-k20-single-target • ratio_twcj_ove | ratio | 1.3058 | 1.3058 | 0% | - | 1 |
| ic:curve25519-k20-single-target • ratio_twcj_ove | ratio | 1.3963 | 1.3963 | 0% | - | 1 |
| ic:curve25519-k20-single-target • ratio_twcj_ove | ratio | 1.3602 | 1.3602 | 0% | - | 1 |
| ic:curve25519-k20-single-target • ratio_twcj_ove | ratio | 0.8312 | 0.8312 | 0% | - | 1 |
| ic:curve25519-k20-single-target • ratio_twcj_ove | ratio | 0.7797 | 0.7797 | 0% | - | 1 |
| ic:curve25519-k20-single-target • s_f1_x07_w01 | S | 2.3781 | 2.3781 | 0% | 1.4287 | 1 |
| ic:curve25519-k20-single-target • s_f1_x07_w0 | S | 2.3211 | 2.3211 | 0% | 1.4951 | 1 |
| ic:curve25519-k20-single-target • s_f1_x07_w1 | S | 2.8907 | 2.8907 | 0% | 1.0818 | 1 |
| ic:curve25519-k20-single-target • s_twcj_w01 | S | 2.1428 | 2.1428 | 0% | 1.3059 | 1 |
| ic:curve25519-k20-single-target • s_twcj_w0 | S | 2.1383 | 2.1383 | 0% | 1.3963 | 1 |
| ic:curve25519-k20-single-target • s_twcj_w1 | S | 2.183 | 2.183 | 0% | 0.8311 | 1 |
| ic:curve25519-k20-single-target • twcj_walk_s | S | 1.3895 | 1.3895 | 0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_f2_over_ | ratio | 1.1475 | 1.1475 | -0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_f2_over_ | ratio | 1.1636 | 1.1636 | -0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_f2_over_ | ratio | 1.0384 | 1.0384 | -0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_tw_over_ | ratio | 0.9801 | 0.9801 | 0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_twcj_ove | ratio | 0.8609 | 0.8609 | 0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_twcj_ove | ratio | 0.8882 | 0.8882 | 0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_twcj_ove | ratio | 0.675 | 0.675 | 0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_twcj_ove | ratio | 1.2961 | 1.2961 | 0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_twcj_ove | ratio | 1.4114 | 1.4114 | 0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_twcj_ove | ratio | 1.3883 | 1.3883 | 0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_twcj_ove | ratio | 0.7492 | 0.7492 | 0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_twcj_ove | ratio | 0.9063 | 0.9063 | 0% | - | 1 |
| ic:curve25519-k24-single-target • s_f1_x07_w01 | S | 1.8945 | 1.8945 | 0% | 1.4975 | 1 |
| ic:curve25519-k24-single-target • s_f1_x07_w0 | S | 1.8341 | 1.8341 | 0% | 1.5802 | 1 |
| ic:curve25519-k24-single-target • s_f1_x07_w1 | S | 2.4383 | 2.4383 | 0% | 1.1058 | 1 |
| ic:curve25519-k24-single-target • s_twcj_w01 | S | 1.6241 | 1.6241 | 0% | 1.2961 | 1 |
| ic:curve25519-k24-single-target • s_twcj_w0 | S | 1.623 | 1.623 | 0% | 1.4113 | 1 |
| ic:curve25519-k24-single-target • s_twcj_w1 | S | 1.6342 | 1.6342 | 0% | 0.7492 | 1 |

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|---|---|---|---|---|---|---|
| ic:curve25519-k24-single-target • twcj_walk_s | S | 1.394 | 1.394 | 0% | - | 1 |
| ic:koblitz-chain-s3-arity-m5m6 • cells_recorded | count | 148 | 148 | - | - | 1 |
| ic:split-prime-p1-rr-dickson-torus • capped_jobs | count | 8 | 2 | 75% | - | 2 |
| ic:split-prime-p1-rr-dickson-torus • rr_attempts | count | 3 | 3 | - | - | 1 |
| ic:split-prime-p1-rr-dickson-torus • rr_degree_r | count | 6 | 6 | 0% | - | 1 |
| ic:split-prime-p1-rr-dickson-torus • rr_vs_f_wid | ratio | 2.29 | 2.29 | 0% | - | 1 |
| ic:split-prime-p1-rr-dickson-torus • rr_vs_t_wid | ratio | 1.33 | 1.33 | 0% | - | 1 |
| ic:split-prime-p1-rr-dickson-torus • t_a2_wall | wall_s | 21,266.672 | 21,266.672 | - | - | 1 |
| ic:split-prime-p1-rr-dickson-torus • t_degree_re | count | 9 | 9 | 0% | - | 2 |
| ic:split-prime-p1-rr-dickson-torus • t_widest_co | count | 1.305e+06 | 1.305e+06 | 0% | - | 1 |
| ic:split-prime-p2-twist-coset-toy • coverage_law | percent | 0.93 | 0.93 | 0% | - | 1 |
| ic:split-prime-p2-twist-coset-toy • f4_work_rel_ | percent | 0.39 | 0.39 | 0% | - | 1 |
| ic:split-prime-p2-twist-coset-toy • f4_work_rel_ | percent | 0 | 0 | - | - | 1 |
| ic:split-prime-p2-twist-coset-toy • gain_min_t4 | ratio | 2.25 | 2.25 | -0% | - | 1 |
| ic:split-prime-p2-twist-coset-toy • purity_nontr | count | 0 | 0 | - | - | 1 |
| ic:split-prime-p2-twist-coset-toy • unplanted_ov | ratio | 0.896 | 0.896 | 0% | - | 1 |
| ic:split-prime-p2-twist-coset-toy • unplanted_ta | count | 0 | 0 | - | - | 1 |
| kernel:c25519_family_bsgs_vs_rho • bsgsk20w0 | S | 0.981 | 0.981 | 0% | 0.6056 | 1 |
| kernel:c25519_family_bsgs_vs_rho • bsgsk24w0 | S | 1.041 | 1.041 | 0% | 0.8852 | 1 |
| kernel:f6_ic_gate • f4_control_word_xors_off_ove | ratio | 1 | 1 | - | - | 1 |
| kernel:f6_ic_gate • f6ic_cpu_off_over_on_max | ratio | 10.29 | 10.29 | -0% | - | 1 |
| kernel:f6_ic_gate • f6ic_cpu_off_over_on_min | ratio | 2.67 | 2.67 | -0% | - | 1 |
| kernel:f6_ic_gate • f6ic_word_xors_off_over_on_m | ratio | 1.84 | 1.84 | -0% | - | 1 |
| kernel:inherited_f4_solve_tree • wall_seq_over_p | ratio | 0.936 | 0.936 | -0% | - | 1 |
| kernel:inherited_f4_solve_tree • wall_seq_over_p | ratio | 2.883 | 2.883 | -0% | - | 1 |
| kernel:inherited_f4_solve_tree • wall_seq_over_p | ratio | 1.589 | 1.589 | -0% | - | 1 |
| kernel:inherited_f4_solve_tree • work_par_over_s | ratio | 2.865 | 2.865 | 0% | - | 1 |
| registry-rebinding:ic-aimed-gather • raw_pins_re | count | 211 | 211 | - | - | 1 |
| rho:koblitz_signed_frobenius_panel • s_legacy_ov | ratio | 0.718 | 0.718 | -0% | - | 1 |
| rho:koblitz_signed_frobenius_panel • s_legacy_ov | ratio | 1.0948 | 1.0948 | -0% | - | 1 |
| rho:koblitz_signed_frobenius_panel • s_legacy_ov | ratio | 1.4723 | 1.4723 | -0% | - | 1 |
| rho:koblitz_signed_frobenius_panel • s_legacy_ov | ratio | 1.0653 | 1.0653 | -0% | - | 1 |
| rho:koblitz_signed_frobenius_panel • s_legacy_ov | ratio | 1.0008 | 1.0008 | -0% | - | 1 |
| rho:prime_medium_p4_confirm • positive_control_r | ratio | 1.4639 | 1.4639 | - | - | 1 |
| rho:prime_medium_p4_confirm • theta_w0 | ratio | 1.3624 | 1.3624 | - | - | 1 |
| rho:prime_medium_tier_bound • pc_bsgs_tb_over_ne | ratio | 1.4893 | 1.4893 | -0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/bsgs- | S | 1.0827 | 1.0827 | 0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/bsgs- | S | 0.9842 | 0.9842 | 0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/bsgs- | S | 1.9684 | 1.9684 | 0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/rho-n | S | 1.0422 | 1.0422 | 0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/rho-n | S | 0.9473 | 0.9473 | 0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/rho-n | S | 1.8961 | 1.8961 | 0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/rho-n | S | 1.0658 | 1.0658 | 0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/rho-n | S | 0.969 | 0.969 | 0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/rho-n | S | 1.9367 | 1.9367 | 0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/rho-p | S | 1.2703 | 1.2703 | 0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/rho-p | S | 1.2703 | 1.2703 | 0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/rho-p | S | 1.2703 | 1.2703 | 0% | - | 1 |
| rho:prime_medium_tier_bound • rho_neg_medium_con | ratio | 1.0809 | 1.0809 | 0% | - | 1 |
| rho:prime_medium_tier_bound • rho_neg_medium_con | ratio | 1.1735 | 1.1735 | 0% | - | 1 |
| rho:prime_medium_tier_bound • rho_neg_medium_con | ratio | 1.067 | 1.067 | 0% | - | 1 |
| rho:prime_medium_tier_bound • rho_neg_medium_con | ratio | 2.1323 | 2.1323 | 0% | - | 1 |
| rho:prime_medium_tier_bound • rho_plain_medium_c | ratio | 1.4868 | 1.4868 | 0% | - | 1 |
| study:elliptic_code_list_decoding_20261107 • d1_ | count | 0 | 0 | - | - | 1 |
| study:minors_guess_determine_20261107 • envelope | fmul | 477,524.385 | 477,524.385 | 0% | 4.3688 | 1 |
| study:minors_guess_determine_20261107 • envelope | fmul | 1.168e+06 | 1.168e+06 | 0% | 6.1114 | 1 |
| study:minors_guess_determine_20261107 • envelope | fmul | 7.096e+06 | 7.096e+06 | 0% | 19.3362 | 1 |
| study:minors_guess_determine_20261107 • envelope | fmul | 8.149e+07 | 8.149e+07 | 0% | 81.8921 | 1 |

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|---|---|---|---|---|---|---|
| study:minors_guess_determine_20261107 • envelope | fmul | 1.27e+09 | 1.27e+09 | 0% | 364.8449 | 1 |
| study:minors_guess_determine_20261107 • envelope | fmul | 1.995e+10 | 1.995e+10 | 0% | 1,551.7891 | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.5155 | 0.5155 | - | - | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.4922 | 0.4922 | - | - | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.4822 | 0.4822 | - | - | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.4863 | 0.4863 | - | - | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.4854 | 0.4854 | - | - | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.4735 | 0.4735 | - | - | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.5019 | 0.5019 | - | - | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.5043 | 0.5043 | - | - | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.4981 | 0.4981 | - | - | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.4992 | 0.4992 | - | - | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.4976 | 0.4976 | - | - | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.4975 | 0.4975 | - | - | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.4991 | 0.4991 | - | - | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.4992 | 0.4992 | - | - | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.4991 | 0.4991 | - | - | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.4993 | 0.4993 | - | - | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.4984 | 0.4984 | - | - | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.4977 | 0.4977 | - | - | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.5001 | 0.5001 | - | - | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.4993 | 0.4993 | - | - | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.5005 | 0.5005 | - | - | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.5002 | 0.5002 | - | - | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.5 | 0.5 | - | - | 1 |
| study:ml_learnability_control_20261107 • heldout | ratio | 0.4998 | 0.4998 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-a • st | fmul | 6.601e+11 | 6.601e+11 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-a • st | count | 0 | 0 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-a • st | fmul | 4.878e+12 | 4.878e+12 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-a • st | count | 0 | 0 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-a • st | count | 0 | 0 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-a • st | wall_s | 12,792 | 12,792 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-a • st | count | 6 | 6 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-a • st | count | 6 | 6 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-a • st | count | 6 | 6 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-a • st | count | 6 | 6 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-a • st | count | 6 | 6 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-a • st | count | 10 | 10 | - | 1 | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-c • st | count | 42 | 42 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-c • st | count | 6 | 6 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-c • st | count | 0 | 0 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-c • st | count | 0 | 0 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-c • st | count | 6 | 6 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-c • st | count | 2 | 2 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-c • st | count | 2 | 2 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-c • st | count | 3 | 3 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-c • st | count | 12 | 12 | - | 1 | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-g3c • | count | 98 | 98 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-g3c • | count | 0 | 0 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-x-t12 | count | 6 | 6 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-x-t12 | count | 2 | 2 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-x-t12 | wall_s | 1,824.0252 | 1,824.0252 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-x-t12 | wall_s | 1,762.8466 | 1,762.8466 | - | - | 1 |
| study:pkm_falldegree_unblock_20261107:pfd-x-t12 | count | 6 | 6 | - | - | 1 |
| study:split_graph_secant_cost_20261107 • cs1_ove | ratio | 1.1495 | 1.1495 | 0% | - | 1 |
| study:split_graph_secant_cost_20261107 • cs1_ove | ratio | 1.3682 | 1.3682 | 0% | - | 1 |
| study:split_graph_secant_cost_20261107 • cs1_ove | ratio | 1.0341 | 1.0341 | 0% | - | 1 |
| study:split_graph_secant_cost_20261107 • cs1_ove | ratio | 1.1567 | 1.1567 | 0% | - | 1 |

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|---|---|---|---|---|---|---|
| study:split_graph_secant_cost_20261107 • cs1_ove | ratio | 0.958 | 0.958 | 0% | - | 1 |
| study:split_graph_secant_cost_20261107 • cs1_ove | ratio | 1.0599 | 1.0599 | 0% | - | 1 |
| study:split_graph_secant_cost_20261107 • cs2_ove | ratio | 3.5921 | 3.5921 | 0% | - | 1 |
| study:split_graph_secant_cost_20261107 • cs2_ove | ratio | 3.6477 | 3.6477 | 0% | - | 1 |
| study:split_graph_secant_cost_20261107 • cs2_ove | ratio | 3.5856 | 3.5856 | 0% | - | 1 |
| study:split_graph_secant_cost_20261107 • cs2_ove | ratio | 3.9811 | 3.9811 | 0% | - | 1 |
| study:split_graph_secant_cost_20261107 • cs2_ove | ratio | 3.6385 | 3.6385 | 0% | - | 1 |
| study:split_graph_secant_cost_20261107 • cs2_ove | ratio | 4.4432 | 4.4432 | 0% | - | 1 |

Every value is quoted from the record it names, in that record’s unit. Percentages and ratios under ’derived’ are computed here from those values and are not measurements. Series in different units are never compared.